



ASR-based segmentation for the analysis of larger child-speech datasets: Performance evaluation on vowels from Australian-English speaking children aged 4 to 11 years

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Abstract

Annotation of segment boundaries in child speech presents a persistent challenge, particularly with large-scale datasets. While auto-segmentation methods have been developed for adult speech, little attention has been paid to evaluating their performance on child speech.

This study evaluates factors contributing to performance of automatic segmentation by analyzing the annotations of two studies into Australian-English-speaking children. The first study assesses human-human reliability in 3- and 12-year-olds. The second compares manual and the Montreal Forced Aligner (MFA) segmentation in 4- to 11-year-olds.

The results indicate that the MFA falls short of the human annotator, though discrepancies decrease as children grow older. Systematic discrepancies between the human annotator and the MFA suggest different criteria for placing segment boundaries. These results shed light on how to incorporate adult-based automatic aligners into semi-automatic (and thus: still partially manual) acoustic analysis of child speech.

Index Terms: auto-segmentation, child speech annotation

1. Introduction

The development of children's speech production is investigated with increasingly large datasets [1]. Traditionally, the first, time-consuming, step of many acoustic analyses of speech is hand annotation of segment boundaries. Substantial time can be saved by automating this step using automatic segmentation methods, a type of automatic speech recognition (ASR) also referred to as forced alignment. These tools are typically trained on adult speech and have been widely tested and successfully implemented to speed up the acoustic analysis of adult speech (see review from [2]). The present study contributes to the evaluation of the feasibility of using such adult-based automatic segmentation to facilitate the analysis of (larger) datasets of speech of children of various ages.

It is known that the performance of automatic segmentation tools is reduced when applied to child speech. There are two commonly used metrics of performance: one is alignment accuracy, measuring the proportion of automatically determined segment intervals that match a true phone; the other is temporal accuracy, measuring the (absolute) temporal distance between the (accurately matched) automatic and manual alignments [3]. Mahr et al. [4] evaluated the performance of popular automatic segmentation algorithms on American-English child speech and reported that the best-performing aligners achieved up to 89% phone-level alignment accuracy—approaching but not reaching human-level precision (96% in their paper). For these accurate alignments, the temporal accuracy in terms of mean boundary-location differences is 26-29 ms. Similarly, Knowles et al.

[3][5] found an alignment accuracy of approximately 88% for vowels in one corpus. For the accurately aligned vowels, the temporal accuracy in terms of absolute boundary-location differences lies around 20 ms. Both studies focused more on alignment than temporal accuracy to evaluate the performance of automatic segmentation tools, finding that factors contributing to suboptimal performance include differences in training data, forced alignment algorithms, child age and segment types. However, for incorporation of forced alignment methods into acoustic child speech research, understanding the impact of these factors on temporal alignment is equally if not more important.

From the perspective of a child speech researcher, it is critical to understand how age contributes to the performance of forced alignment algorithms. Overall, previous results show that automatic speech segmentation is better as children are older [3][4]. It is well established that children's speech undergoes rapid changes throughout development [6][7], contributing to significant between-child variability [8]. And even the speech within a single child at one age will display substantial variability, due to developing motor control and phonological proficiency [8]. This variability in children's speech can persist even into adolescence and cause difficulty for ASR systems such as automatic transcription [9][10][11]. However, previous evaluation studies on automatic segmentation have only explored a limited age range (from 2 to 7 years of age). The current study aims to expand this scope by exploring the effect of age on automatic segmentation performance in a wider range (until 11 years of age).

Previous evaluation studies have also found that the alignment of vowels tends to be more accurate than consonants [3][4]. While consonants were separated out into subtypes (e.g., stops, sibilants, in [3]), all vowels were grouped together. However, children's production of weak vowels specifically changes over the course of development. Compared with adult speech, children tend to produce longer and stronger schwa, for example in articles [12][13]. Moreover, young children's production of schwa is highly variable [14]. Therefore, this study aims to compare the performance of automatic segmentation on strong versus weak vowels.

The previous evaluations of automatic segmentation in child speech have omitted a further important factor for the annotation of segment boundaries—boundary type, the onset versus offset of a segment. Even with exceedingly clear segmentation protocols and human coders, mean inter-labeler deviations between two human coders are smaller at the start than the end of a vowel [15]. For example, Wesenick & Kipp [16] find the mean deviation of plosive-vowel boundaries (i.e., the start of the vowel) to be 6 ms, but the deviation in vowel-plosive boundaries (i.e., the end of the vowel) to be 12 ms. However, only

one study has assessed the role of boundary type on automatic segmentation performance, reporting inconsistent results across two different corpora [3]. Our study presents a further step in this direction by comparing vowel onsets with offsets.

Overall, to address these gaps in the literature, this paper thus explores the performance of an automatic segmentation tool on child speech, with a focus on child age, vowel types, and boundary type.

2. Methods

This study builds on the previously made annotations for two studies into how Australian-English speaking children incorporate articles into their prosodic structure. Study 1 uses the annotations from [13] of 3 and 12 year old children. A subset of data have been coded by two human coders to establish Human-Human reliability. Study 2 uses the annotations from an in-progress study into 4-to-11 year old children. All data were forced aligned by the Montreal Forced Aligner (MFA) [17] and then manually corrected by a human coder, to establish Human-ASR reliability.

2.1. Corpus

The data utilized for both studies were sourced from AusKidTalk, a corpus comprising audio recordings of children's speech in Australian-English [18]. Both studies focus exclusively on the sentence-repetition task in this corpus, where each child was instructed to listen to and repeat each of the 36 pre-recorded sentences.

All the raw data were diarized by the AusKidTalk group [19], serving as the first step to identify suitable recordings for research and isolate children's utterances from surrounding speech with corresponding orthographic transcription.

2.2. Participants

Study 1 is based on a finished study [13] which analyses data from 32 Australian English-speaking children aged 3 and 12 years. The speech of 40% of the children was coded by two independent annotators. Eventually, 8 children contributed to Study 1 with 4 children aged 3 (2 boys, 2 girls) and 4 children aged 12 (4 males, 4 females).

Study 2 is based on an in-progress study which includes speech data from 110 children aged 4 to 11 (55 males, 55 females).

2.3. Materials

The studies are based on two experimental conditions from the AusKidTalk sentence-repetition task, called the "No-Clitic" and "Clitic" conditions. All stimuli elicited Subject-Verb-Object sentences of 5 to 6 syllables. The difference between these conditions is the presence of an article in the Clitic condition, for example *Boys often cut the cards*, and the absence of an article in the No-Clitic condition, for example *Boys often cut cards*. Following the research goals of Studies 1 and 2, we focused on a limited number of vowels: two monophthong vowels /e,ʊ/ and one diphthong /aɪ/ in the verbs and schwa /ə/ in the article.

In Study 1, the total number of utterance tokens considered was 96 for the 3-year-old group and 96 for the 12-year-old group. Study 2 encompassed vowel 5051 tokens across all ages (4-to-11 years old). The tokens coded as an error by annotators (see below, section workflow) were excluded from analysis, resulting in 152 and 4343 included tokens in Studies 1 and 2,

respectively.

2.4. Workflow

Both Study 1 and Study 2 employed a semi-automated workflow designed for the acoustic analysis of large-scale child speech datasets. Differences in the workflow are due to Study 2, including use of a forced aligner.

2.4.1. Study 1: Manual workflow to assess Human-Human inter-annotator agreement

For Study 1, two human annotators independently annotated 40% of the dataset from two age groups (3-year-old and 12-year-old) to assess Human-Human reliability without the use of forced alignment tools. Segment boundaries were manually placed by two annotators. The workflow was conducted in Praat [20] and involved two steps:

First, annotators identified mismatches between the target and children's production. Such errors were categorized into phonological, morphological, and fluency-related error types, including (among others) the error tags: "NR" (Not realized by Children), "OM" (when children omitted an article, the format of tag will be "OM-the") or "SC" (self correction).

Then, annotators manually segmented target vowels in children's speech, placing segment boundaries based on waveform periodicity and second formant (F2) visibility in spectrograms. In cases of spectral smearing, waveform data was prioritized for boundary placement.

2.4.2. Study 2: Semi-automatic workflow to assess Human-ASR agreement

Study 2 required a semi-automated approach due to the larger dataset size. For study 2, one expert annotator from study 1 and one forced aligner segmented the speech data from children aged 4 to 11 years to assess Human-ASR agreement.

Forced alignment was conducted using the Montreal Forced Aligner (MFA) [17], as this showed the best performance in child speech datasets across different dialects of English [4][5][21]. As no pre-trained model for Australian English exists within MFA, we used the US-English model. The workflow included:

First, the human annotator coded error labels, as for Study 1. The utterances without error labels were fed into MFA to generate automatic alignments in a Praat TextGrid. To facilitate manual correction, a Praat script was used to convert the MFA intervals into points on a tier for the onsets and offsets of the verb and article vowel targets. Finally, the annotator manually adjusted these points.

2.5. Evaluation

During the annotation process, we observed that in the elicited task recording environment, children's speech frequently overlapped with either adult speech or a beep sound used as a cue to signal the beginning of an utterance. This overlap often resulted in the forced aligner producing entirely mis-aligned utterances rather than temporal errors at the phoneme level. Following [3], the present study thus separately measured two types of segmentation accuracy: alignment accuracy and temporal accuracy.

2.5.1. Alignment accuracy

Alignment accuracy for Study 2 was defined as the proportion of automatically generated segment intervals that included the midpoint of the corresponding manually determined segment intervals. Segment intervals meeting this criterion were coded as matched alignments, and all others as unmatched alignments. Note that unmatched cases were not observed for the Human-Human comparison of Study 1.

2.5.2. Temporal accuracy

Temporal accuracy for Study 2 was assessed to provide a more detailed evaluation of matched alignments by measuring the absolute differences between the matched automatically and manually placed boundaries. Temporal accuracy was also measured for Study 1 between two human annotators to establish a baseline for human annotation consistency. Both Human-Human and Human-ASR temporal accuracy were calculated based on absolute boundary-location differences.

Boundary-location differences were converted into accuracy rate, with absolute differences below 25 ms representing acceptable discrepancies, and larger discrepancies representing disagreement. The 25 ms is the gold-standard threshold used to evaluate forced aligners and has been implemented in acoustic research to evaluate both inter-labeler and Human-ASR reliabilities [17][22].

2.5.3. Statistical model

Alignment accuracy was analyzed using a logistic mixed-effects model, with age included as a fixed effect and a random intercept specified for each participant. Age was normalized using a centering and scaling transformation.

For the Human-Human temporal accuracy in Study 1, a limited number of data points was available, so no statistical model was constructed. Instead, absolute boundary-location differences and the accuracy rates were reported as descriptive statistics.

For model analysis of Human-ASR temporal accuracy, boundary-location differences (non-absolute), serving as the first measure of temporal accuracy, was analyzed with a linear mixed-effects model. Boundary-location accuracy, the second, binary measure of temporal accuracy, was fitted by a logistic-mixed effects regression model. Both models included fixed effects for age, boundary types (onset coded as -0.5 contrast with offset coded as +0.5), and vowel type (strong as -0.5, weak as +0.5), with random intercepts for individual children. More complex random structures were not included due to issues with non-convergence and singularity warnings

3. Result

3.1. Alignment accuracy

The logistic mixed-effects model on accuracy shows no significant effect of age (point estimate of the odds ratio = 8.01; 95% CI from 0.29 to 218.92; p from 1 = 0.209). Consequently, no definitive conclusion can be drawn regarding the effect of children's age on the alignment accuracy of forced aligners.

3.2. Temporal accuracy

3.2.1. Age

The descriptive results of absolute boundary-location differences and accuracies are shown in Table 1. In the Human-

Human condition, the average absolute boundary-location differences are 6.63 ms (variance = 0.098) for age 3 and 7.28 ms (variance = 0.101) for age 12. The corresponding accuracy rates (relative to the 25 ms threshold) are 92.9% and 96.8%, respectively, indicating a similar level of agreement between human annotators as observed in comparable previous studies [4].

In contrast, the Human-ASR comparison exhibits larger absolute differences across all ages, with mean values ranging from 12.02 ms to 15.44 ms. The highest discrepancy is observed at (the youngest) age 4 (mean = 15.44 ms, variance = 0.21), while the lowest discrepancy occurs at age 10 (mean = 12.02 ms, variance = 0.097), closely followed by (the oldest) age 11 (mean = 12.21 ms, variance = 0.1 ms). These results are mirrored in the accuracy rates, with the lowest recorded at age 4 (88.7%), and the highest at age 10 (94.2%), followed by age 11 (93.7%). These descriptive match those of previous Human-ASR comparisons for child speech [3].

Table 1: Absolute boundary-duration differences and percentages of acceptable discrepancies

	Age	Number of tokens	Absolute differences (ms) Mean (Var)	Accuracies (%)
Human-Human	3	96	6.63 (0.098)	92.9
	12	96	7.28 (0.101)	96.8
Human-ASR	4	168	15.44 (0.21)	88.7
	5	418	13.04 (0.17)	91.8
	6	269	14.96 (0.2)	91.0
	7	351	13.53 (0.18)	91.2
	8	300	13.06 (0.15)	92.3
	9	335	13.04 (0.12)	93.7
	10	413	12.02 (0.097)	94.2
	11	334	12.21 (0.1)	93.7

The (non-absolute) boundary-location differences displayed in Figure 1 show on average negative values, indicating that the automatic segmentation typically placed boundaries earlier than the human. The linear mixed-effects model confirms this with a significantly negative intercept (point estimate = -5.87 ms; 95% CI from -6.58 to -5.15; $p = 2 \times 10^{-16}$).

The linear mixed-effects model on the boundary-location differences indicates a negative slope for age (point estimate = -4.49 ms; 95% CI from -6.87 to -2.1 ms; p from 0 = 0.00028). The logistic regression model on accuracy shows a similar pattern, with accuracy increasing with age (point estimate of the odds ratio = 1.99; CI from 0.94 to 4.26; p from 1 = 0.074). Thus, the performance of the automatic aligner is better (i.e., more similar to the human) as children are older.

3.2.2. Vowel type

Figure 1 plots the boundary-duration differences as a function of age for strong and weak vowels across two prosodic boundary types. The top row in Figure 1 indicates that the weak vowels exhibited larger boundary-location differences than the full vowels, which is confirmed by the significant effect of vowel type in the linear mixed-effects model (point estimate = 9.98 ms; CI from 8.69 ms to 11.27 ms; $p = 2 \times 10^{-16}$) but not the logistic mixed-effects model on temporal accuracy (point estimate of the odds ratio = 0.95; CI from 0.64 to 1.39; p from 1 = 0.78). This shows that automatically segmented weak vowels require more hand-correction than strong vowels, particularly under stricter acceptance criteria.

There is also a significant interaction effect between age and vowel type in both models: the extent to which discrepancies were larger in our weak than our strong vowels was larger in the speech of younger than older children (linear model: by

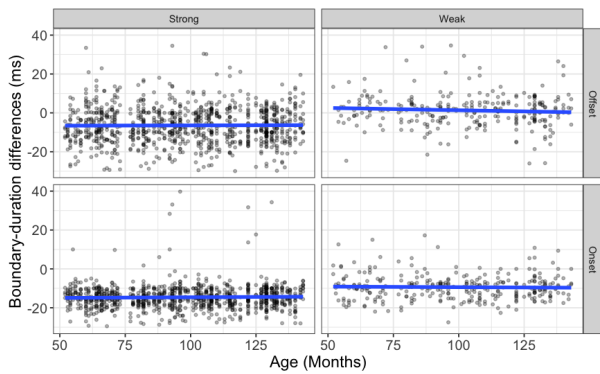


Figure 1: Age-related changes in boundary-duration differences (ms) across vowel type (strong vs. weak) and boundary type (onset vs. offset). Grey dots represent individual data points; blue lines indicate linear regression trends.

8.58 ms, CI = 4.22 ms to 12.94 ms, $p = 0.00011$; logistic model: point estimate of the odds ratio = 4.62; CI from 1.21 to 17.59; p from 1 = 0.03). Thus, the Human-ASR temporal discrepancies between the two types of vowels narrow as children are older.

3.2.3. Boundary type

The bottom row in Figure 1 indicates that the vowel offsets exhibited larger boundary-location differences than the vowel onsets, which is confirmed by the significant effect of boundary type in the linear mixed-effects model on the differences (point estimate = 12.85 ms; CI from 11.56 to 14.14 ms; $p = 2 \times 10^{-16}$) and the logistic mixed-effects model on the accuracy (point estimate of the odds ratio = 0.5; CI from 0.34 to 0.74; $p = 0.0005$). These results indicate that vowel offsets need more hand correction than vowel onsets.

There is also a significant interaction effect between age and boundary type: the extent to which discrepancies were larger in our offset than our onset vowel boundaries was larger in younger than older children's speech (point estimate = 8.58 ms; CI from 4.22 ms to 12.94 ms; $p = 6.32 \times 10^{-7}$) and (point estimate of the odds ratio = 4.94; CI from 1.3 to 18.76; p from 1 = 0.018). The boundary-location differences between the two types of boundaries thus narrow as children are older.

3.2.4. Further interactions

Although not directly anticipated, the remaining two interactions were also significant in one or both models. There is a significant interaction effect between vowel type and boundary type in both models on boundary-location differences: the difficulties with offsets (compared to onsets) are larger in weak vowels than strong vowel in the linear model (point estimate = 11.08 ms; CI from 8.51 ms to 13.65 ms; $p = 2 \times 10^{-16}$) and in the logistic model on accuracy (point estimate of the odds ratio = 0.35; CI from 0.16 to 0.76; p from 1 = 0.007). Finally, there is a significant three way interaction between age, verb type and boundary type in the linear model, indicating that the extent to which the difficulties with offsets are larger in weak than strong vowel became smaller as children grow older (point estimate = -20.52 ms; CI from -29.22 ms to -11.82 ms; $p = 0.000039$). Note that this interaction effect is not significant in the logistic model on accuracy (point estimate of the odds ratio = 10.1; CI from 0.7 to 145.13; p from 1 = 0.089).

4. Discussion

The objective of this set of studies was to assess the performance of automatic segmentation, specifically the MFA [17], in terms of its alignment and temporal accuracy for child speech (ages 4 to 11 years), comparing strong versus weak vowels as well as segment onsets and offsets.

The results of alignment accuracy highlight the importance of manual checks following automatic (forced) alignment. While no significant effect of age was found, unmatched alignments occurred across all age groups, indicating that the automatic aligner did not always approximate the approximately correct interval for a segment. Moreover, the reported alignment accuracy is contingent and substantially improved by the automatic speaker diarization and manual error coding in our workflow. Informal inspection of forced alignment results in the absence of diarization and error coding suggest that alignment accuracy would deteriorate without these steps. Therefore, implementing a semi-automatic (rather than fully automatic) workflow with manual steps before and after automatic alignment is thus still critical when analyzing child speech.

The results on temporal accuracy demonstrate superior performance for the Human-Human alignment (7 ms on average) compared to the Human-ASR alignment (approximately 13.5 ms on average), suggesting that automatic segmentation still falls short of human precision. Systematic discrepancies between the human annotator and the MFA may partially account for this difference, as the MFA consistently places boundaries earlier than the human annotator. Considering just vowel onsets, this may reflect that human annotators consistently refer to formant structure stabilization to mark the beginning of vowels, thus ignoring much of the variability in children's realizations. The MFA, on the other hand, may use a multitude of cues, for example marking some vowel onsets during the initial periodicity before a vowel takes its full spectral form.

Regarding age effects on the temporal accuracy of the automatic aligner, the present study found that discrepancies between human annotators and the ASR decreased, while alignment accuracy improved as children grew older. This is consistent with the previous evaluation studies [3][4], reinforcing the notion that state-of-the-art adult-based automatic segmentation systems struggle with processing younger children's speech. The results from this study furthermore highlight that manual correction for younger children requires particular attention where automatic alignment performance is less reliable—specifically, for weak vowels and offset boundaries.

5. Conclusion

In this study, we assessed how age, vowel type and boundary affect the performance of the Montreal Forced Aligner on child speech of 4-to 11-year-old children. The results indicate that the MFA falls short of the human annotator, though discrepancies decrease as children grow older. Systematic discrepancies between the human annotator and the MFA suggest different criteria for placing segment boundaries. These results shed light on how to incorporate adult-based automatic aligners into semi-automatic (and thus: still partially manual) acoustic analysis of child speech.

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7. References

- [1] S. Stoll and R. Schikowski, "Child-language corpora," in *A practical handbook of corpus linguistics*. Springer, 2021, pp. 305–327.
- [2] J. Yuan, W. Lai, C. Cieri, and M. Liberman, "Using forced alignment for phonetics research," in *Chinese Language Resources: Data Collection, Linguistic Analysis, Annotation and Language Processing*. Springer, 2023, pp. 289–301.
- [3] T. Knowles, M. Clayards, and M. Sonderegger, "Examining factors influencing the viability of automatic acoustic analysis of child speech," *Journal of Speech, Language, and Hearing Research*, vol. 61, no. 10, pp. 2487–2501, 2018.
- [4] T. J. Mahr, V. Berisha, K. Kawabata, J. Liss, and K. C. Hustad, "Performance of forced-alignment algorithms on children's speech," *Journal of Speech, Language, and Hearing Research*, vol. 64, no. 6S, pp. 2213–2222, 2021.
- [5] T. Knowles, M. Clayards, M. Sonderegger, M. Wagner, A. Nadig, and K. H. Onishi, "Automatic forced alignment on child speech: Directions for improvement," in *Proceedings of Meetings on Acoustics*, vol. 25, no. 1. AIP Publishing, 2015.
- [6] S. Lee, A. Potamianos, and S. Narayanan, "Acoustics of children's speech: Developmental changes of temporal and spectral parameters," *The Journal of the Acoustical Society of America*, vol. 105, no. 3, pp. 1455–1468, 1999.
- [7] S. Gautam and L. Singh, "Developmental changes of spectral parameter in children speech," in *2016 3rd International Conference on Signal Processing and Integrated Networks (SPIN)*. IEEE, 2016, pp. 220–225.
- [8] M. E. Beckman, A. R. Plummer, B. Munson, and P. F. Reidy, "Methods for eliciting, annotating, and analyzing databases for child speech development," *Computer speech & language*, vol. 45, pp. 278–299, 2017.
- [9] M. Gerosa, D. Giuliani, S. Narayanan, and A. Potamianos, "A review of asr technologies for children's speech," in *Proceedings of the 2nd Workshop on Child, Computer and Interaction*, 2009, pp. 1–8.
- [10] M. Gerosa, D. Giuliani, and F. Brugnara, "Acoustic variability and automatic recognition of children's speech," *Speech Communication*, vol. 49, no. 10–11, pp. 847–860, 2007.
- [11] A. Potamianos and S. Narayanan, "Robust recognition of children's speech," *IEEE Transactions on speech and audio processing*, vol. 11, no. 6, pp. 603–616, 2003.
- [12] L. Gerken, "Prosodic structure in young children's language production," *Language*, pp. 683–712, 1996.
- [13] R. Cai, P. Boersma, I. Yuen, K. Demuth, and T. Benders, "Prosodic clitics in english-speaking children's speech production—an acoustic study," in *Proc. SpeechProsody 2024*, 2024, pp. 56–60.
- [14] M. A. Redford, "Grammatical word production across metrical contexts in school-aged children's and adults' speech," *Journal of Speech, Language, and Hearing Research*, vol. 61, no. 6, pp. 1339–1354, 2018.
- [15] P. Machač and R. Skarnitzl, *Principles of phonetic segmentation*. Epocha, 2009.
- [16] M.-B. Wesenick and A. Kipp, "Estimating the quality of phonetic transcriptions and segmentations of speech signals," in *Proceeding of Fourth International Conference on Spoken Language Processing. ICSLP'96*, vol. 1. IEEE, 1996, pp. 129–132.
- [17] M. McAuliffe, M. Socolof, S. Mihuc, M. Wagner, and M. Sonderegger, "Montreal forced aligner: Trainable text-speech alignment using kaldi," in *Interspeech*, vol. 2017, 2017, pp. 498–502.
- [18] L. Rumberg, C. Gebauer, H. Ehlert, M. Wallbaum, L. Bornholt, J. Ostermann, and U. Lüdtkke, "kidstalc: A corpus of 3-to 11-year-old german children's connected natural speech," in *INTER-SPEECH*, 2022, pp. 5160–5164.
- [19] T. O. Szalay, M. Shahin, T. Sirojan, Z. Nan, R. Huang, J. Arciuli, E. Baker, F. Cox, K. J. Ballard, and B. Ahmed, "Auskidtalk: Developing an orthographic annotation workflow for a speech corpus of australian english-speaking children," *Available at SSRN 5020250*.
- [20] P. Boersma and D. Weenink, "Praat: doing phonetics by computer [computer program]," 2025, version 6.4.27, retrieved 27 January 2025 from <http://www.praat.org/>.
- [21] S. Gonzalez, J. Grama, and C. E. Travis, "Comparing the performance of forced aligners used in sociophonetic research," *Linguistics Vanguard*, vol. 6, no. 1, p. 20190058, 2020.
- [22] H. Wu, J. Yun, X. Li, H. Huang, and C. Liu, "Using a forced aligner for prosody research," *Humanities and Social Sciences Communications*, vol. 10, no. 1, pp. 1–13, 2023.